King's College London

University Of London

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PLACE this paper and any answer booklets **in the EXAM ENVELOPE provided**

Candidate No: ..... **Desk No:** .....

MSc Examination

7CCMFM04 Stochastic Analysis Exam

January 2026

Time Allowed: Two Hours

All questions carry equal marks. Full marks will be awarded for complete answers to all FOUR questions.

Within a given question, the relative weights of the different parts are indicated by a percentage figure.

NO calculators are permitted.

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1. Throughout this question, we let $W = (W_t)_{t\geq 0}$ be a real-valued Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

a. State the definition of a Lévy process. **[30%]**

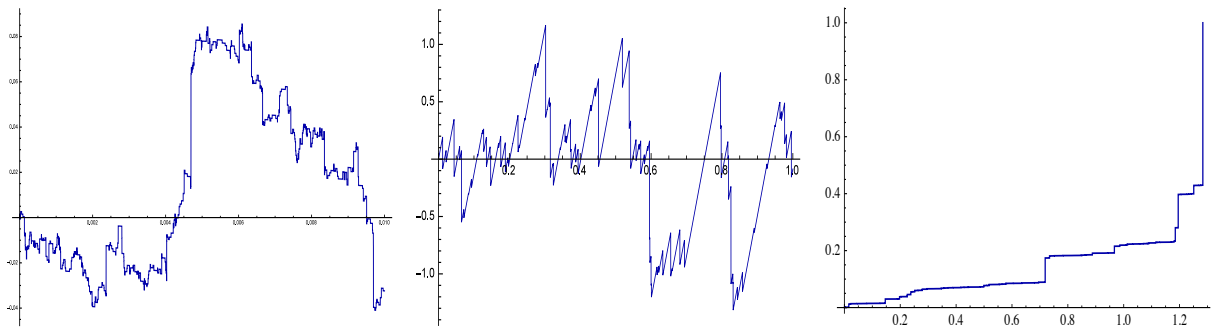
Solution. X has independent increments, $X_t - X_s \overset{(d)}{=} X_{t-s}$ (stationary increments) and for all $\varepsilon > 0$, $\lim_{h\to 0} \mathbb{P}(|X_{t+h} - X_t| > \varepsilon) = 0$ (continuity in probability).

b. State the definition of a Gaussian process. **[30%]**

Solution. For any $n \in \mathbb{N}$ and ordered sequence $t_1 < ... < t_n$, $(X_{t_1}, ..., X_{t_n})$ has a multivariate Normal distribution

c. Which of the processes plotted below is a compound Poisson process, a CGMY process, and an inverse hitting time process $(\tau_b)_{b\geq 0}$ for Brownian motion? **[40%]**

Solution. 1st plot is CGMY, 2nd is compound Poisson, third is τ_b



2. Throughout this question, we let $W = (W_t)_{t\geq 0}$ be a real-valued Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and $\tau_b = \inf\{t \geq 0 : W_t = b\}$ for $b \geq 0$.

a. Give an example of a right-continuous process X for which the running supremum is not attained, and draw a picture. **[30%]**

Solution. Example: let $X_t = \mu t - N_t$ with $\mu > 0$, where N is a Poisson process and let T_1 be the first jump time of N. Then

$$\sup_{0\leq t\leq T_1} X_t \quad = \quad X_{T_1-} \quad = \quad \mu T_1$$

but this sup is not attained because $X_{T_1} = \mu T_1 - 1$, due to right continuity of N.

b. Let $X_t = \mu t + W_t$ and $\bar{X}_t = \sup_{0\leq s\leq t} X_s$. Using that

$$\mathbb{P}(\bar{X}_t > b \,|\, X_t = x) \quad = \quad e^{-h(b,x)/t} \tag{1}$$

for $b \geq \max(x, 0)$ where $h(b, x) = 2b(b - x)$, write down an integral expression for $\mathbb{P}(\bar{X}_t > b)$. (see also Hwk1 q2, which also uses Eq 1). **[30%]**

Solution. Clearly $X_t \geq b$ implies that $\bar{X}_t \geq b$. Using this, and the tower property of conditional expectation, we see that

$$\begin{aligned}
\mathbb{P}(\bar{X}_t > b) \quad &= \quad \mathbb{E}(1_{\bar{X}_t>b}) \quad = \quad \mathbb{E}(\mathbb{E}(1_{\bar{X}_t>b}|X_t)) \quad = \quad \mathbb{E}(\mathbb{P}(\bar{X}_t > b|X_t)) \\
&= \quad \int_{-\infty}^{b} \mathbb{P}(\bar{X}_t > b|X_t = x)\,\mathbb{P}(X_t \in dx) \;+\; \mathbb{P}(X_t \geq b) \\
&= \quad \int_{-\infty}^{b} e^{-h(b,x)/t}\, p_t^{(\mu)}(x)dx \;+\; \Phi^c(\frac{b-\mu t}{\sqrt{t}})
\end{aligned}$$

where $p_t^{(\mu)}(x) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-\mu t)^2}{2t}}$ is the density of X_t. We can use this formula to price One-Touch options under the Black-Scholes model.

c. State the general form of the Lévy-Khintchine formula for the mgf of a Lévy process X. Show that the log mgf of X_1 is convex. **[30%]**

Solution. $\mathbb{E}(e^{pX_t}) = e^{V(p)t}$, where

$$V(p) \quad = \quad \mu p + \frac{1}{2}\sigma^2 p^2 \;+\; \int_{-\infty}^{\infty} (e^{px} - 1 - px1_{|x|\leq 1})\nu(dx)$$

for some μ, σ and $\nu(.) \geq 0$. We now re-write the integral term as

$$\int_{-1}^{1} (e^{px} - 1 - px)\nu(dx) \;+\; \int_{[-1,1]^c} (e^{px} - 1)\nu(dx)\,.$$

Then (assuming we can take derivatives inside the integral sign) we see that

$$V''(p) \quad = \quad \sigma^2 \;+\; \int_{-1}^{1} x^2 e^{px}\nu(dx) \;+\; \int_{[-1,1]^c} x^2 e^{px}\nu(dx) \quad = \quad \sigma^2 \;+\; \int_{-\infty}^{\infty} x^2 e^{px}\nu(dx) \quad \geq \quad 0$$

which verifies convexity.

3. Throughout this question, we let $W = (W_t)_{t\geq 0}$ be a real-valued Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

a. Let X be a CGMY-type process with drift and negative-only jumps with $\mathbb{E}(e^{pX_t}) = e^{tV(p)}$, where

$$V(p) \quad = \quad \mu p + (G+p)^Y - G^Y$$

for $p \geq -G$, where $G > 0$ and $Y \in (1,2)$. Explain how to compute $\mathbb{E}(e^{-qH_b})$ for $q > 0$ by solving a certain equation, where $H_b := \inf\{t \geq 0 : X_t = b\}$ for $b \geq 0$ (proof not required). **[30%]**

Solution. From Eq 2 in q5, Hwk 4, $\mathbb{E}(e^{-qH_b}) = e^{-\Phi(q)b}$, where $\Phi(q)$ is the largest root of $V(p) = q$. The easy way to remember this is to just use that $\mathbb{E}(e^{pX_t - V(p)t}) = 1$, and then assume we can replace t with H_b and set $V(p) = q$ (this requires a limiting argument to justify properly, again see q5,Hwk 4) (note the equation $V(p) = q$ can only solved explicitly here if $\mu = 0$ as in Hwk4,q6, in which case there is only 1 root given by $\Phi(q) = (q + G^Y)^{\frac{1}{Y}} - G$).

b. For what range of μ-values is $\mathbb{P}(H_b < \infty) < 1$, i.e. $\mathbb{P}(H_b = \infty) > 0$? You may use that $V(p) = (G^{Y-1}Y + \mu)p + O(p^2)$ as $p \to 0$ and that $V''(p) > 0$ for $p \geq -G$. **[30%]**

Solution. $\mathbb{E}(e^{-qH_b}) = \mathbb{E}(e^{-qH_b}1_{H_b<\infty})$, so

$$\lim_{q\to 0}\mathbb{E}(e^{-qH_b}) \quad = \quad \lim_{q\to 0}\mathbb{E}(e^{-qH_b}1_{H_b<\infty}) \quad = \quad \mathbb{E}(1_{H_b<\infty}) \quad = \quad \mathbb{P}(H_b < \infty) \quad = \quad e^{-b\Phi(0+)}$$

where $\Phi(0+) = \lim_{q\searrow 0}\Phi(q)$. From a picture, we see that $\Phi(q)$ does **not** tend to zero as $q \to 0$ if and only if $V'(0) < 0$, i.e. if $G^{Y-1}Y + \mu < 0$.

c. Compute the distribution of $\bar{X}_{e_q}$ and $\bar{X}_\infty$, where $\bar{X}_t = \sup_{0\leq s\leq t} X_s$ and $e_q \sim \text{Exp}(q)$ is independent of X. How would you compute the density of H_b. **[20%]**

Solution. Again from q5,Hwk4, we know that $\bar{X}_{e_q} \sim \text{Exp}(\Phi(q))$. For the second part, $e_q \to \infty$ a.s. as $q \to 0$ (see final sentence of the solution), so $\bar{X}_\infty \sim \text{Exp}(\lim_{q\to 0}\Phi(q)) = \text{Exp}(\Phi(0+))$, if we are in the case where $\mathbb{P}(H_b < \infty) = \mathbb{P}(\bar{X}_\infty > b) < 1$, where the limit here is positive. Otherwise $\bar{X}_\infty = \infty$ a.s.

If $E \sim \text{Exp}(1)$ then $\mathbb{P}(E/q > x) = \mathbb{P}(E > qx) = e^{-qx} \to 1$ as $q \to 0$. Clearly $E/q \to \infty$ a.s., and the previous sentence shows that $\frac{1}{q}E \sim \text{Exp}(q)$, so $e_q \to \infty$.

Recall the hitting time density can be obtained via an IFT as

$$f_{H_b}(t) \quad = \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-iut}\mathbb{E}(e^{iuH_b})du$$

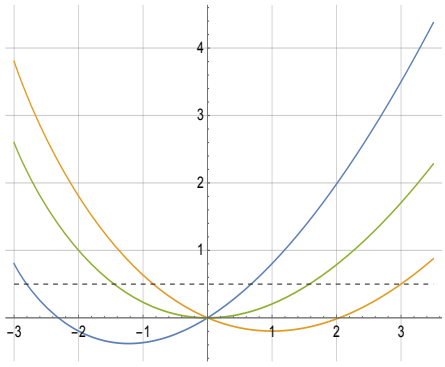


Figure 1: Plot of $V(p)$ for $G = 3$, $Y = \frac{3}{2}$ with $\mu = -2$ (blue), $\mu = -2.598$ (green) and $\mu = -4$ (yellow), and $q = \frac{1}{2}$ here.

d. If $S_t = S_0 e^{X_t}$ is a stock price process and $\mathbb{P}$ is a risk-neutral measure, what condition is required for the model to be arbitrage-free, i.e. for $\tilde{S}_t = e^{-rt} S_t$ to be a martingale?

Solution. **[20%]**

$$e^{-rt}\mathbb{E}(S_t) \quad = \quad e^{-rt} S_0 e^{V(1)t} \quad = \quad S_0 e^{t(-r+\mu+(G+1)^Y - G^Y)}$$

so the no-arbitrage condition is $-r + \mu + (G+1)^Y - G^Y = 0$.

4. **a.** Let X be a compound Poisson process with Lévy density $\nu(x) = ce^{-\rho x}1_{x>0}$ for $c, \rho > 0$. What is the jump rate for X? **[20%]**

Solution. $\nu(x) = \lambda\mu(x)$, where $\mu(x) = \rho e^{-\rho x}1_{x>0}$ is the density of an $\text{Exp}(\rho)$ random variable and $\lambda = c/\rho$. Hence λ is the rate of jumps.

b. Let X be sfBM, which is a zero-mean stationary Gaussian process with autocovariance $r(\tau) = \mathbb{E}(X_t X_{t+\tau}) = (T^{2H} - |\tau|^{2H})^+$ for $H \in (0, \frac{1}{2})$, $\tau \in \mathbb{R}$ and $T > 0$. Apply the Kolmogorov continuity theorem to X on the interval $[0, T]$. **[30%]**

Solution. Assume that $0 \le s \le t \le T$ and set $\tau = t - s$. Then using that $X_t \sim X_0$ for all t (i.e. stationarity),

$$\begin{aligned}
\mathbb{E}((X_t - X_s)^2) &= \mathbb{E}(X_t^2) - 2\mathbb{E}(X_s X_t) + \mathbb{E}(X_s^2) &&= 2\mathbb{E}(X_0^2) - 2\mathbb{E}(X_{t-s}X_0) \\
&&&= 2(r(0) - r(\tau)) \\
&&&= 2(T^{2H} - (T^{2H} - \tau^{2H})) \quad = \quad 2|t-s|^{2H}.
\end{aligned}$$

But (aside from the pre-factor 2) this is the same answer we get for fBM, so we then use the same argument with the Kolmogorov continuity theorem to show that X is $H - \varepsilon$-Hölder continuous since X is also Gaussian. **[30%]**

c. Consider a **double exponential Kou** compound Poisson process with

$$\nu(x) = a\rho e^{-\rho x}1_{x>0} + \hat{a}\hat{\rho}e^{\hat{\rho}x}1_{x<0}$$

with $\rho, \hat{\rho} > 0$, i.e. where the jump sizes are drawn from a two-sided asymmetric exponential distribution. Compute $\mathbb{E}(\bar{X}_{e_q})$, where $e_q \sim \text{Exp}(q)$ is independent of X. You may use that (from the Lévy-Khintchine formula) $\mathbb{E}(e^{zX_t}) = e^{t\psi(z)}$ where

$$\psi(z) = \int_{-\infty}^{\infty} (e^{zx} - 1)\nu(x)dx = \frac{az}{\rho - z} - \frac{\hat{a}z}{\hat{\rho} + z}$$

for $\hat{\rho} < z < \rho$, and infinity otherwise. Is $\mathbb{P}(\bar{X}_{e_q} = 0) = 0$?

Solution. From Hwk 4 q7 we know that

$$\mathbb{E}(e^{z\bar{X}_{e_q}}) = \frac{1 - \frac{z}{\rho}}{1 - \frac{z}{\zeta(q)}} \tag{2}$$

where $\zeta(q) \in (0, \rho)$ is the positive root of $q - \psi(z) = 0$. Then

$$\mathbb{E}(\bar{X}_{e_q}) = \frac{d}{dz}\mathbb{E}(e^{z\bar{X}_{e_q}})|_{z=0} = \frac{1}{\zeta(q)} - \frac{1}{\rho} > 0.$$

For the final part, the answer is no, since X is a compound Poisson process, and hence may not jump at all over $[0, e_q]$. In fact, the distribution which has Eq (2) as its mgf has $\bar{X}_{e_q} = 0$ with probability $\frac{\zeta}{\rho}$ and $\bar{X}_{e_q} \sim \text{Exp}(\zeta)$ with probability $1 - \frac{\zeta}{\rho}$, so $\bar{X}$ has a discrete and continuous component.